MODULE 36 · EXAM STRATEGY - ALL DOMAINS

Practice Exam 1

Twenty questions, weighted by domain, worked in full. Take it under time.

QUESTIONS 20 SUGGESTED TIME 40 minutes PASS GUIDE 14 of 20

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Before you start

- **Set a timer for 40 minutes.** Twenty questions at two minutes each is the real exam's pace, and pace is a skill worth practising.
- **Write your answers down** before watching the workings. Deciding is worth several times more than recognising.
- Each question pauses for nine seconds. **Pause it properly** if you need longer, and note anything you guessed.
- **14 of 20 is a reasonable pass indication.** It is not the real scaled score, which is compensatory and calculated differently.

> ON THE EXAM
>
> Mark the questions you guessed even when you got them right. Those are the ones to revisit, because luck does not repeat on the day.

An operations team must alarm when available memory on a fleet of EC2 instances falls below 10 percent. CloudWatch shows CPU, network and disk metrics for the instances but no memory metric of any kind.

The instances have an instance profile and are managed by Systems Manager.

What must the team do first?

A Enable detailed monitoring on the instances

B Install and configure the CloudWatch agent to publish memory metrics

C Create a metric filter on the instances' system logs

D Enable high-resolution metrics on the default EC2 namespace

◆ PAUSE AND DECIDE

What must the team do first?

A **Detailed monitoring**

Changes the interval from five minutes to one for metrics that already exist. It cannot create a metric the hypervisor cannot observe.

B **Install the CloudWatch agent**

EC2 metrics come from the hypervisor, which cannot see inside the guest. Memory and disk space usage require the agent.

C **Metric filter on system logs**

Would work only if something were already writing memory figures to a log group. Nothing here suggests that.

D **High-resolution EC2 metrics**

Resolution and existence are different axes. A metric that is not published cannot be published more often.

THE SIGNAL

Resolution, interval and existence are three different things. If a metric does not exist, no setting will summon it.

An alarm on a batch processing queue is configured to fire when a custom metric exceeds a threshold. The metric is published only while the batch job runs, roughly twice a day. The alarm spends most of its time in INSUFFICIENT_DATA and occasionally fires spuriously overnight.

The team wants the alarm to reflect only the periods when the job is actually running.

What should the team configure?

A Treat missing data as breaching

B Reduce the evaluation period to one minute

C Switch the alarm to anomaly detection

D Treat missing data as missing, so gaps neither trigger nor clear the alarm

◆ PAUSE AND DECIDE

What should the team configure?

A Missing as breaching

Would make the alarm fire whenever the job is not running, which is most of the day. That is the opposite of the requirement.

B Shorter period

Evaluates the same absent data more often. A shorter period on a metric that is not being published changes nothing.

C Anomaly detection

Learns a normal band from history. A metric absent for twenty-two hours a day gives it almost nothing to learn from.

D Missing as missing

Gaps are ignored entirely: they neither trigger nor clear the alarm, so its state reflects only the periods when data exists.

THE SIGNAL

Four missing-data treatments exist because silence means different things. Here silence means idle, so it should mean nothing at all.

> A security team must be alerted within minutes whenever anyone calls DeleteTrail in any account. They also need to answer, on demand, how many times a particular API was called by a specific role over the last month.
>
> CloudTrail delivers to a CloudWatch log group and to S3.

Which combination meets both requirements?

A. A metric filter with an alarm for the alerting, and Logs Insights or Athena for the on-demand question

B. Logs Insights queries on a schedule for both

C. A subscription filter to Lambda for both

D. CloudTrail Insights for both

◆ PAUSE AND DECIDE

Which combination meets both requirements?

A Metric filter plus a query tool

Two requirements with opposite shapes: forward-looking alerting is a metric filter and alarm; a backward-looking question is a query over stored data.

B Scheduled queries for both

Queries answer the second requirement well. Polling on a schedule is a poor alerting mechanism and adds latency and cost.

C Subscription filter for both

Streams matching events to a consumer, which could alert. It does not answer historical questions about data already stored.

D CloudTrail Insights

Detects unusual rates of API activity. It is not a targeted alert on a specific call, nor an ad hoc query tool.

THE SIGNAL

A metric filter is forward-looking and creates a metric from events as they arrive. Logs Insights is backward-looking and queries what is already stored.

> An engineer cannot connect to an EC2 instance in a private subnet using Session Manager. The instance appears in the EC2 console as running, has the SSM agent installed and a suitable instance profile attached, but does not appear as a managed node in Systems Manager.
>
> The subnet has no route to a NAT gateway or internet gateway.

What is the most likely cause?

A The instance profile lacks ssm:StartSession

B Session Manager requires inbound port 22 to be open

C The instance has no network path to the Systems Manager endpoints, and needs interface endpoints for ssm, ssmmessages and ec2messages

D The SSM agent version is too old to support Session Manager

◆ PAUSE AND DECIDE

What is the most likely cause?

A Missing ssm:StartSession

That permission belongs to the human starting the session, not to the instance profile. It also would not stop the node registering.

B Needs inbound 22

Confidently wrong, and the opposite of the point. Session Manager requires no inbound ports at all, which is why it replaces bastions.

C No network path to the SSM endpoints

Registration requires three prerequisites: agent, permissions, and a network path. Two are stated as present and the third is explicitly absent.

D Agent too old

Possible in principle, and the scenario supplies a much more specific cause. A stated absent route beats a speculative version problem.

THE SIGNAL

Three prerequisites: agent, permissions, network path. The scenario confirms two and then rules out the third in its final sentence.

> A data processing job on an m6i.4xlarge reads from a 6 TiB gp3 volume, provisioned at 50,000 IOPS and 1,000 MiB/s. For the first half hour of each run it achieves close to 1,000 MiB/s. It then settles at roughly half that, for the remaining hours.
>
> The volume's BurstBalance metric reports no data.

What is the cause?

A The gp3 volume's burst credits are exhausted

B The provisioned throughput exceeds what a 6 TiB volume supports

C The file system needs a larger block size

D The instance's EBS bandwidth burst has expired, dropping it to its sustained baseline

◆ PAUSE AND DECIDE

What is the cause?

A gp3 burst credits

gp3 has no burst mechanism at all, which is exactly why BurstBalance reports no data. That absence disproves this option.

B Throughput exceeds the volume

A 6 TiB gp3 volume comfortably supports 1,000 MiB/s, and the run achieves it for the first half hour, which proves the volume can deliver it.

C Block size

Might improve efficiency generally. It would not produce a clean halving exactly half an hour into every run.

D Instance EBS burst expired

Smaller instance sizes hold their maximum EBS bandwidth for around 30 minutes in 24 hours, then drop to roughly half. Both the timing and the ratio match.

THE SIGNAL

Two numbers point at the instance: about 30 minutes, and about half. The missing BurstBalance metric independently rules out the volume.

A newly deployed Auto Scaling group behind an ALB never reaches its desired capacity. Instances launch, are marked unhealthy after about two minutes, are terminated, and are replaced. The application takes around five minutes to start and pass its health check.

The group uses ELB health checks.

What should be changed?

A Increase the health check grace period beyond the application's start-up time

B Increase the instance warm-up to 300 seconds

C Switch the group to EC2 health checks

D Increase the target group's unhealthy threshold

◆ PAUSE AND DECIDE

What should be changed?

A Increase the grace period

The grace period is precisely how long after launch before health checks may terminate an instance. Set below start-up time, it condemns every instance.

B Instance warm-up

Warm-up excludes a booting instance from the group's scaling metrics. It never prevents a termination, which is what is happening here.

C EC2 health checks

Stops the loop by no longer checking whether the application serves traffic. It fixes the symptom by removing the monitoring.

D Unhealthy threshold

Delays the verdict by a few checks. Against a three-minute gap between judgement and readiness, that is not enough.

THE SIGNAL

Launch, terminate, repeat, forever, has exactly one cause. Only one of the four timers offered can terminate an instance at all.

> Users of an application behind an Application Load Balancer intermittently receive HTTP 504 responses. The targets are registered and healthy, and the application logs show the affected requests completing successfully after about 75 seconds.
>
> The business confirms these requests legitimately take that long.

What is the immediate cause?

A. No healthy targets are registered in the target group

B. The targets are returning malformed responses

C. The ALB's idle timeout, which defaults to 60 seconds, is shorter than the request duration

D. The deregistration delay is too short

◆ PAUSE AND DECIDE

What is the immediate cause?

A **No healthy targets**

That produces 503, not 504, and the scenario states the targets are registered and healthy.

B **Malformed responses**

That produces 502. The logs show requests completing successfully, so the response is well formed, merely late.

C **Idle timeout shorter than the request**

504 means the target did not respond in time. The default 60-second idle timeout against a 75-second request produces exactly this.

D **Deregistration delay**

Governs how long a removed target keeps draining connections. It has no bearing on the timeout of a live request.

THE SIGNAL

502 means someone answered badly. 503 means nobody answered. 504 means nobody answered in time. That sentence separates all three.

> An Aurora PostgreSQL cluster runs a single writer instance with no reader instances. During an Availability Zone failure the database was unavailable for several minutes. The team is confused, because Aurora stores six copies of the data across three Availability Zones.
>
> They require recovery within one minute.

What should the team do?

A Enable Multi-AZ on the Aurora cluster

B Add at least one reader instance in another Availability Zone, so there is something to promote

C Convert the cluster to Aurora serverless

D Reduce the cluster endpoint's TTL

◆ PAUSE AND DECIDE

What should the team do?

A **Enable Multi-AZ**

The team's premise is correct: the storage already spans three AZs. Durability was never the problem.

B **Add a reader instance**

Aurora's fast failover works by promoting an existing replica. With none, AWS must create a new instance, which takes minutes rather than seconds.

C **Aurora serverless**

Changes how capacity scales, not how failover works. A serverless cluster with a single instance has the same problem.

D **Lower the endpoint TTL**

The cluster endpoint follows failover automatically. The delay was creating a writer, not resolving to one.

THE SIGNAL

Durable is not the same as available. The data was safe throughout; there was simply no compute able to serve it.

A company must be able to recover a customer-facing application in a second Region. The agreed RPO is 15 minutes and the agreed RTO is 45 minutes. Cost must be kept as low as those targets allow. A scaled-down copy of the environment is not currently running in the second Region.

Data is replicated continuously; the compute layer exists only as AMIs and an Auto Scaling group set to zero.

Which DR strategy does this describe?

A Backup and restore

B Warm standby

C Active/active

D Pilot light

◆ PAUSE AND DECIDE

Which DR strategy does this describe?

A **Backup and restore**

Nothing runs in the DR Region and data is not continuously replicated. Here the data layer is live, which is a stronger position.

B **Warm standby**

Requires a scaled-down but running environment. The scenario states explicitly that no such copy is running.

C **Active/active**

Both Regions serve production traffic. Nothing in the second Region is serving anything.

D **Pilot light**

The data layer is live and replicating while compute exists but is switched off. That is the definition, and it fits the 45-minute RTO.

THE SIGNAL

The tiebreaker: is anything running and serving in the second Region? No means pilot light. Yes means warm standby.

A CloudFormation stack manages a production RDS database with DeletionPolicy set to Retain. An engineer changes a property that requires replacement of the DB instance. After the update completes, the original database no longer exists.

The team believed Retain protected it.

What should have been set as well?

A A stack policy denying all updates to the database

B DeletionPolicy: Snapshot instead of Retain

C UpdateReplacePolicy: Retain, which governs the old resource when an update replaces it

D Termination protection on the stack

◆ PAUSE AND DECIDE

What should have been set as well?

A **Stack policy**

Genuinely would have blocked the update, which is a different control. The question asks what governs the old resource once the update proceeds.

B **DeletionPolicy: Snapshot**

Still DeletionPolicy, so it fires on the wrong event. A snapshot is also not the same as retaining the database.

C **UpdateReplacePolicy: Retain**

DeletionPolicy fires when a resource leaves the stack. UpdateReplacePolicy fires when an update replaces it, and it defaults to Delete.

D **Termination protection**

Prevents the stack being deleted. The stack was not deleted; a resource within it was replaced.

THE SIGNAL

Two attributes, two different events, both defaulting to Delete. Set both on anything holding data. This is the most expensive gap in CloudFormation.

> An organisation must deploy a baseline set of IAM roles and a Config recorder to every account. New accounts are created most weeks and must be covered without anyone remembering. The organisation uses AWS Organizations with all features enabled.
>
> Deployment must halt if more than three accounts fail.

What should be implemented?

A. A service-managed StackSet targeting the organisation's OUs, with automatic deployment and a failure tolerance of 3

B. A self-managed StackSet listing every account ID

C. A pipeline that deploys a CloudFormation stack into each account

D. An SCP requiring the roles to exist

◆ PAUSE AND DECIDE

What should be implemented?

A Service-managed StackSet on OUs

Automatic deployment covers accounts joining a targeted OU later, and failure tolerance implements the stated halting condition exactly.

B Self-managed with account IDs

Works on the day it is built, and requires someone to add every new account each week. That is the manual step the requirement forbids.

C A pipeline per account

Rebuilds StackSets by hand, with no failure tolerance mechanism and a growing list of deployments to maintain.

D An SCP

SCPs restrict what principals may do. They never create resources, so nothing would be deployed at all.

THE SIGNAL

"Covered without anyone remembering" is automatic deployment, which exists only in the service-managed permission model.

> An ECS service on Fargate fails to start. Tasks move from PROVISIONING to STOPPED with the message 'ResourceInitializationError: unable to pull secrets or registry auth'. The application code is unchanged and the container image exists in ECR.
>
> The task definition specifies a task role and a task execution role.

Which role should be investigated?

A The task role, which the application uses to call AWS APIs

B The task execution role, which ECS uses to pull the image and fetch secrets before the container runs

C The cluster's EC2 instance profile

D The ECS service-linked role

◆ PAUSE AND DECIDE

Which role should be investigated?

A The task role

The application's own identity, used only once the container is running. This failure occurs before any application code executes.

B The task execution role

Pulling the image, fetching secrets and writing logs all happen under the execution role, before the container starts. The error names exactly those actions.

C Instance profile

Fargate has no instances, so there is no instance profile to hold permissions.

D Service-linked role

Lets ECS manage load balancers and scaling on your behalf. It is not used to pull images or retrieve secrets.

THE SIGNAL

The stage of failure identifies the role. Anything happening before your code runs belongs to the execution role.

> During a promotional campaign a Lambda function behind API Gateway returns throttling errors for about two minutes. CloudWatch shows ConcurrentExecutions peaking at roughly 350 against an account limit of 1,000, with a matching spike in Throttles. The concurrency graph rises in distinct steps.
>
> No other function in the account was active.

What is the cause?

A The account concurrency limit was reached

B Reserved concurrency on the function is set too low

C The function's timeout is too short

D The function hit Lambda's scaling rate, which limits how fast new execution environments are created

◆ PAUSE AND DECIDE

What is the cause?

A **Account limit reached**

350 against a limit of 1,000, with nothing else running, means the pool was around 65 percent free.

B **Reserved concurrency too low**

Would produce a flat ceiling at a fixed number, not a rising staircase. Nothing indicates a reservation was set.

C **Timeout too short**

Produces timeout errors in the function's own logs, not throttling at admission. A different failure entirely.

D **The scaling rate**

Lambda creates new execution environments at a fixed rate. A sharp spike hits that ceiling regardless of unused pool capacity, producing the stair-step shape.

THE SIGNAL

Two independent limits exist and only one is on the dashboard people watch. Throttled with capacity free is always the scaling rate.

> An application role has an identity policy allowing s3:GetObject on a bucket and kms:Decrypt on the customer managed key that encrypts it. Downloads fail with AccessDenied. The role and the key are in the same account, and there are no SCPs.
>
> A security engineer recently removed the default statement from the key policy.

What is the cause?

A The key policy no longer delegates authority to IAM, so identity policies cannot grant access to the key

B The role also needs kms:Encrypt

C The bucket needs a bucket policy allowing the role

D The key must be a multi-Region key

◆ PAUSE AND DECIDE

What is the cause?

A Key policy no longer delegates to IAM

A KMS key policy is mandatory and is the ultimate authority. The default statement granting the account root full access is what lets identity policies grant key access.

B Needs kms:Encrypt

Downloading decrypts. Encrypt is needed for writing, and adding it changes nothing here.

C Needs a bucket policy

Same-account access works from the identity policy alone. A bucket policy is not required.

D Multi-Region key

Matters when ciphertext must be decrypted in another Region. Everything here is in one account and one Region.

THE SIGNAL

A KMS key policy is unlike other resource policies: without delegation, IAM grants for that key are simply ignored.

A security team attaches an SCP at the organisation root denying all actions outside three approved Regions. Testing in member accounts confirms it works. A month later an audit finds EC2 instances running in an unapproved Region.

Those instances are in the organisation's management account.

What explains this?

A The SCP must also be attached to each OU

B The root user is exempt from SCPs

C SCPs do not apply to the management account

D SCPs take effect only after the next account refresh

◆ PAUSE AND DECIDE

What explains this?

A Attach to each OU

A policy at the root is inherited by every OU and account beneath it. Re-attaching changes nothing.

B Root user is exempt

Precisely backwards, and the most attractive distractor. SCPs apply to every principal in a member account including its root user.

C SCPs do not apply to the management account

Documented behaviour, and the main reason AWS advises keeping workloads out of the management account entirely.

D Account refresh

Invented behaviour. The scenario also states the policy was confirmed working in member accounts.

THE SIGNAL

The last line of the scenario is the whole answer. Option B is the same fact inverted, which makes it the hardest distractor here.

A compliance team must report across 50 accounts: which resources are shared with accounts outside the organisation, and which IAM roles have been unused for 90 days. They have already enabled an organisation-level external access analyzer.

No unused access findings are appearing.

What is required?

A Enable Security Hub, which aggregates all Access Analyzer finding types

B Create a separate organisation-level unused access analyzer, because each finding type requires its own analyzer

C Wait, because unused access findings require 90 days of history to accumulate

D Enable AWS Config in all accounts, which computes unused access

◆ PAUSE AND DECIDE

What is required?

A Security Hub aggregates them

Security Hub aggregates findings that exist. It cannot produce findings from an analyzer that was never created.

B A separate unused access analyzer

IAM Access Analyzer is three distinct analyzer types: external access, unused access and internal access. Each must be created explicitly.

C Wait for history

The tracking period is configurable and analysis uses existing CloudTrail history. It does not require a fresh 90 days.

D AWS Config

Evaluates resource configuration against rules. It does not analyse access patterns or compute unused permissions.

THE SIGNAL

One service name covering several distinct products. Having one analyzer produces none of the others' findings, and the pricing differs too.

> Servers in an on-premises data centre, connected over Direct Connect, must reach an S3 bucket without traversing the public internet. A gateway endpoint for S3 has been created and added to the VPC's route tables. Instances inside the VPC reach S3 through it correctly.
>
> On-premises traffic still cannot reach S3 privately.

What should the team do?

A Create an interface endpoint for S3, which has a private IP reachable from on-premises

B Add the on-premises CIDR to the gateway endpoint policy

C Advertise the S3 prefix list to on-premises over BGP

D Attach the gateway endpoint to the Direct Connect gateway

◆ PAUSE AND DECIDE

What should the team do?

A Interface endpoint

Gateway endpoints cannot be reached from on-premises over Direct Connect or VPN. An interface endpoint has a private IP in your subnet, and can be.

B Endpoint policy

An endpoint policy restricts what may be done through an endpoint that is already reachable. It cannot make an unreachable one reachable.

C Advertise the prefix list

A gateway endpoint works through a VPC route table entry. There is no path to advertise to an on-premises router.

D Attach to the DX gateway

Gateway endpoints attach to VPC route tables. No such construct exists on a Direct Connect gateway.

THE SIGNAL

The scenario supplies a controlled comparison: in-VPC access works, on-premises does not. That is the documented boundary of a gateway endpoint.

A custom network ACL is applied to a subnet. Clients can establish connections to a web server on port 443, and every response times out. The security groups allow all outbound traffic, and the NACL has an inbound rule allowing TCP 443 from 0.0.0.0/0.

No other rules have been added.

What is missing?

A An inbound security group rule for port 443

B An outbound NACL rule allowing the ephemeral port range, because NACLs are stateless

C An outbound NACL rule allowing TCP 443

D A route to the internet gateway

◆ PAUSE AND DECIDE

What is missing?

A. Inbound SG rule for 443

Connections are being established, which proves the security group already permits the traffic.

B. Outbound ephemeral range

The response leaves from port 443 addressed to the client's high ephemeral port. A stateless NACL needs an outbound rule covering that range.

C. Outbound rule for 443

The near miss. The rule matches on destination port, and the response is destined for the ephemeral port, not for 443.

D. Route to the IGW

The request arrived, so inbound routing works. A missing route would prevent connection entirely rather than hanging the response.

THE SIGNAL

Connects then hangs is the stateless signature. Option C is the trap for anyone who knows a rule is needed but matches on the wrong port.

A multiplayer game backend uses UDP on a custom port across three Regions. Players report high latency, and recovery is slow when a Region becomes unhealthy. Some corporate customers require fixed IP addresses to add to their firewalls.

The team currently uses Route 53 latency-based routing with health checks.

What should the team implement?

A. A CloudFront distribution in front of the game servers

B. Route 53 geoproximity routing with a bias

C. AWS Global Accelerator with endpoint groups in each Region

D. A Network Load Balancer with targets in all three Regions

◆ PAUSE AND DECIDE

What should the team implement?

A CloudFront

HTTP and HTTPS only, and game traffic is not cacheable. It cannot carry UDP at all.

B Geoproximity routing

Still DNS. Failover waits for health checks, then the TTL, then resolver caching, which is the slow recovery being complained about.

C Global Accelerator

UDP on any port, two static anycast IPs for allow-listing, traffic onto the AWS backbone at the nearest edge, and failover in about 30 seconds with no DNS.

D NLB across Regions

An NLB is a Regional resource. It provides neither global anycast entry nor Regional failover.

THE SIGNAL

Three clauses point the same way: UDP eliminates CloudFront, fixed IPs eliminate DNS-based answers, and slow recovery eliminates anything with a TTL.

> A software vendor must expose a single API privately to 150 customer AWS accounts. Many customers use the 10.0.0.0/16 range, which is identical to the vendor's own VPC CIDR. Customers must not be able to reach anything else in the vendor's environment.
>
> The vendor does not want to manage per-customer network connections.

What should the vendor implement?

A VPC peering with each customer account

B A Transit Gateway shared with customers through AWS RAM

C A public API endpoint restricted by an IP allow-list

D A PrivateLink endpoint service behind a Network Load Balancer, with customers creating interface endpoints

◆ PAUSE AND DECIDE

What should the vendor implement?

A Peering per customer

Fails on overlapping CIDRs alone, and 150 peering connections is exactly the per-customer management the vendor ruled out.

B Shared Transit Gateway

Creates routing between vendor and customers, so overlaps break it, and customers gain a route into the vendor's network.

C Public endpoint with allow-list

Traffic traverses the internet, which the requirement excludes, and an allow-list for 150 customers is unmaintainable.

D PrivateLink endpoint service

PrivateLink establishes no routing, so identical CIDRs are irrelevant. It exposes exactly one service, and scales to any number of consumers.

THE SIGNAL

Overlapping CIDRs eliminates every routing-based option at once. One clause removes three of the four answers before any other consideration.

How to use your score

- **14 or more** suggests you are on track. **11 to 13** means revisit the weaker domains before the next practice exam.
- **Note which domain each miss came from.** Five questions were Domain 1, four Domain 2, four Domain 3, three Domain 4, four Domain 5.
- **Review anything you guessed correctly** with the same care as a miss. That is the single highest-value revision you can do.
- **Do not memorise these questions.** The exam will not repeat them; the reasoning patterns are what transfer.

> ON THE EXAM
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> Several questions here shared a shape: a stated fact that eliminates an option, and a final sentence carrying the deciding constraint. Look for both.